Class 9 Candy Mini Project

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This mini-project will allow us to apply our data analysis skills and build on our PCA skills.

```
candy_file <- "candy-data.csv"

candy = read.csv(candy_file, row.names=1)
# head(candy)
```

> Q1. How many different candy types are in this dataset?

```
nrow(candy)
```

```
## [1] 85
```

> Q2. How many fruity candy types are in the dataset?

```
sum(candy$fruity)
```

```
## [1] 38
```

`winpercent` is the percentage of people who prefer a given candy over another randomly chosen candy from the dataset. Higher values indicate a more popular candy.

We can find the winpercent value for Twix by using its name to access the corresponding row of the dataset.

```
candy["Twix", ]$winpercent
```

```
## [1] 81.64291
```

Alternatively, you can do this…

```
# install.packages("vctrs")
# install.packages("dplyr")

library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag
```

```
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
candy |>
  filter(row.names(candy)=="Twix") |>
  select(winpercent)
```

```
##      winpercent
## Twix   81.64291
```

> Q3. What is your favorite candy (other than Twix) in the dataset and what is it's winpercent value? *My favorite candy is Snickers*

```
# This is the winpercent value of the Snickers
candy["Snickers", ]$winpercent
```

```
## [1] 76.67378
```

> Q4. What is the winpercent value for "Kit Kat"?

```
candy["Kit Kat", ]$winpercent
```

```
## [1] 76.7686
```

> Q5. What is the winpercent value for "Tootsie Roll Snack Bars"?

```
candy["Tootsie Roll Snack Bars", ]$winpercent
```

```
## [1] 49.6535
```

There is a useful skim() function in the skimr package that can help give you a quick overview of a given dataset.

```
# install.packages("skimr")
library("skimr")
skim(candy)
```

Data summary

Name	candy
Number of rows	85
Number of columns	12
_____	
Column type frequency:	
numeric	12
_____	
Group variables	None

Variable type: numeric

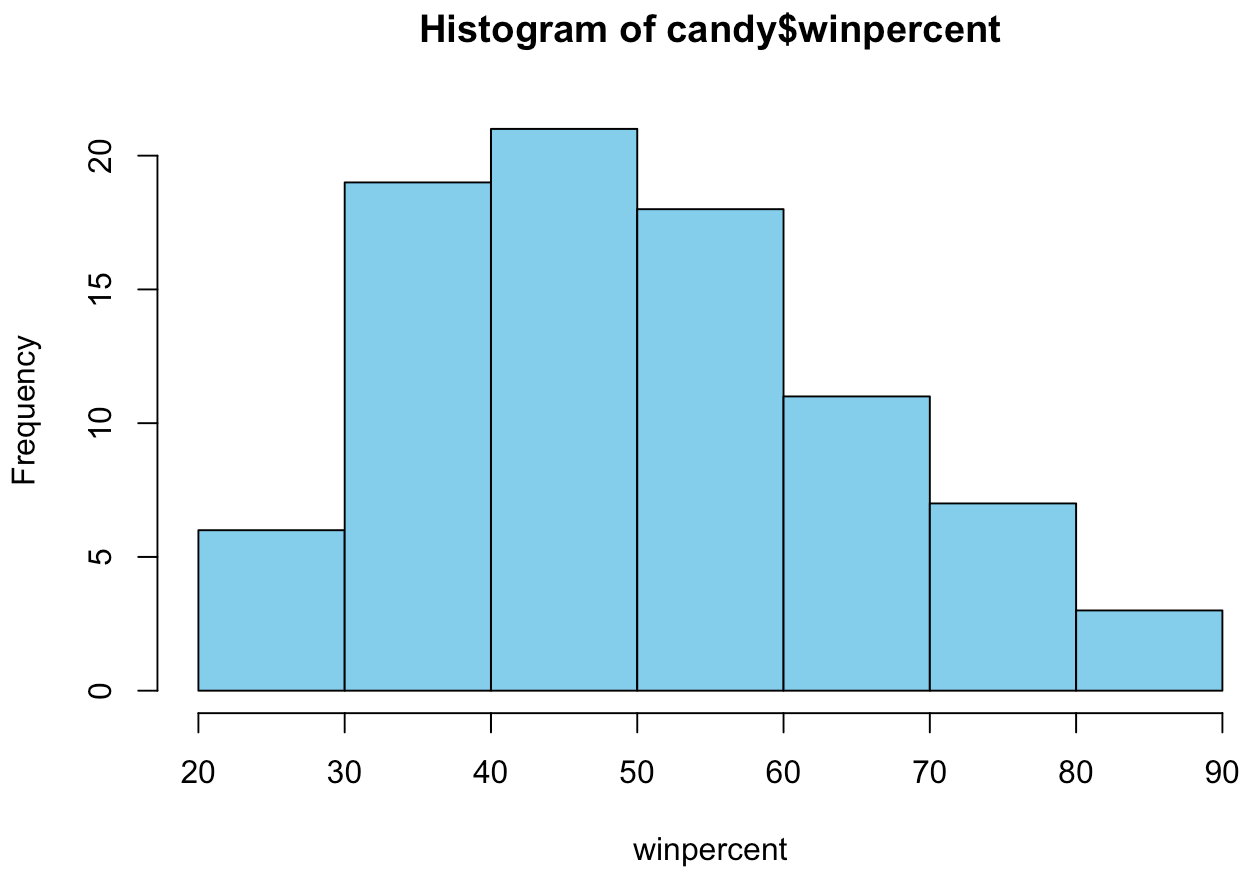
skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyalmondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedricewafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

Q6. Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset? *The column with the types of candy are different because it's not a numerical scale*

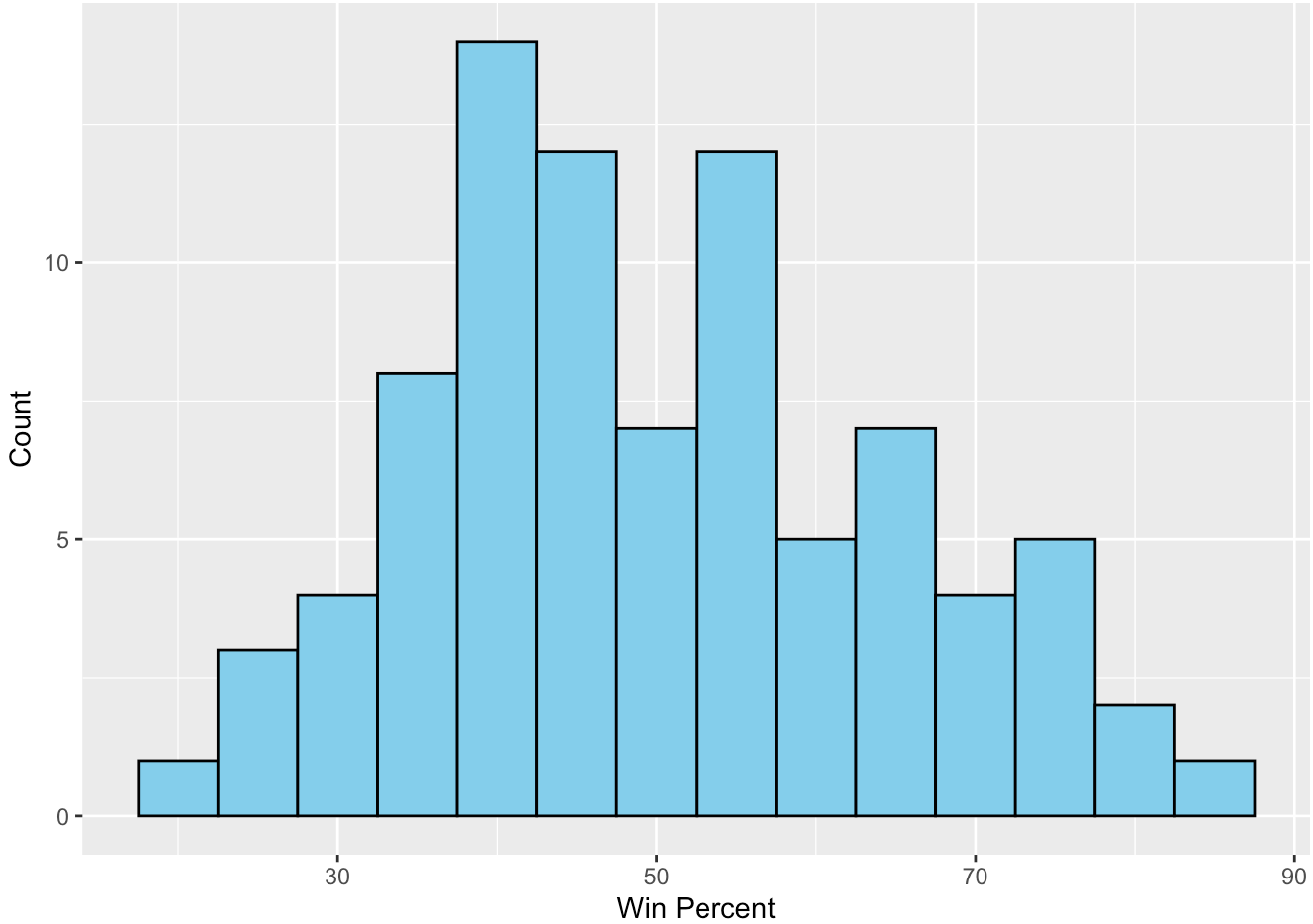
Q7. What do you think a zero and one represent for the candy$chocolate column? *The zero means that there isn't chocolate in that specific candy, and a one means that there is choclate in that specific candy*

Q8. Plot a histogram of winpercent values using both base R an ggplot2.

```
hist(candy$winpercent, xlab = "winpercent", col = "skyblue")
```




```
# install.packages("ggplot2")
library(ggplot2)
ggplot(candy, aes(x = winpercent)) +
  geom_histogram( binwidth = 5, fill = "skyblue", color = "black") +
  labs( x = "Win Percent",
       y = "Count")
```



Q9. Is the distribution of winpercent values symmetrical? *They are not symmetrical*

Q10. Is the center of the distribution above or below 50%?

```
mean(candy$winpercent)
```

```
## [1] 50.31676
```

```
median(candy$winpercent)
```

```
## [1] 47.82975
```

Q11. On average is chocolate candy higher or lower ranked than fruit candy?

```
avg_chocolate <- mean(candy$winpercent[as.logical(candy$chocolate)])
avg_fruity <- mean(candy$winpercent[as.logical(candy$fruity)])
avg_chocolate > avg_fruity
```

```
## [1] TRUE
```

Q12. Is this difference statistically significant?

```
t.test(
  candy$winpercent[as.logical(candy$chocolate)],
  candy$winpercent[as.logical(candy$fruity)]
)
```

```
##
##  Welch Two Sample t-test
##
## data:  candy$winpercent[as.logical(candy$chocolate)] and candy$winpercent[as.logical
(candy$fruity)]
## t = 6.2582, df = 68.882, p-value = 2.871e-08
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  11.44563 22.15795
## sample estimates:
## mean of x mean of y
##  60.92153  44.11974
```

Q13. What are the five least liked candy types in this set?

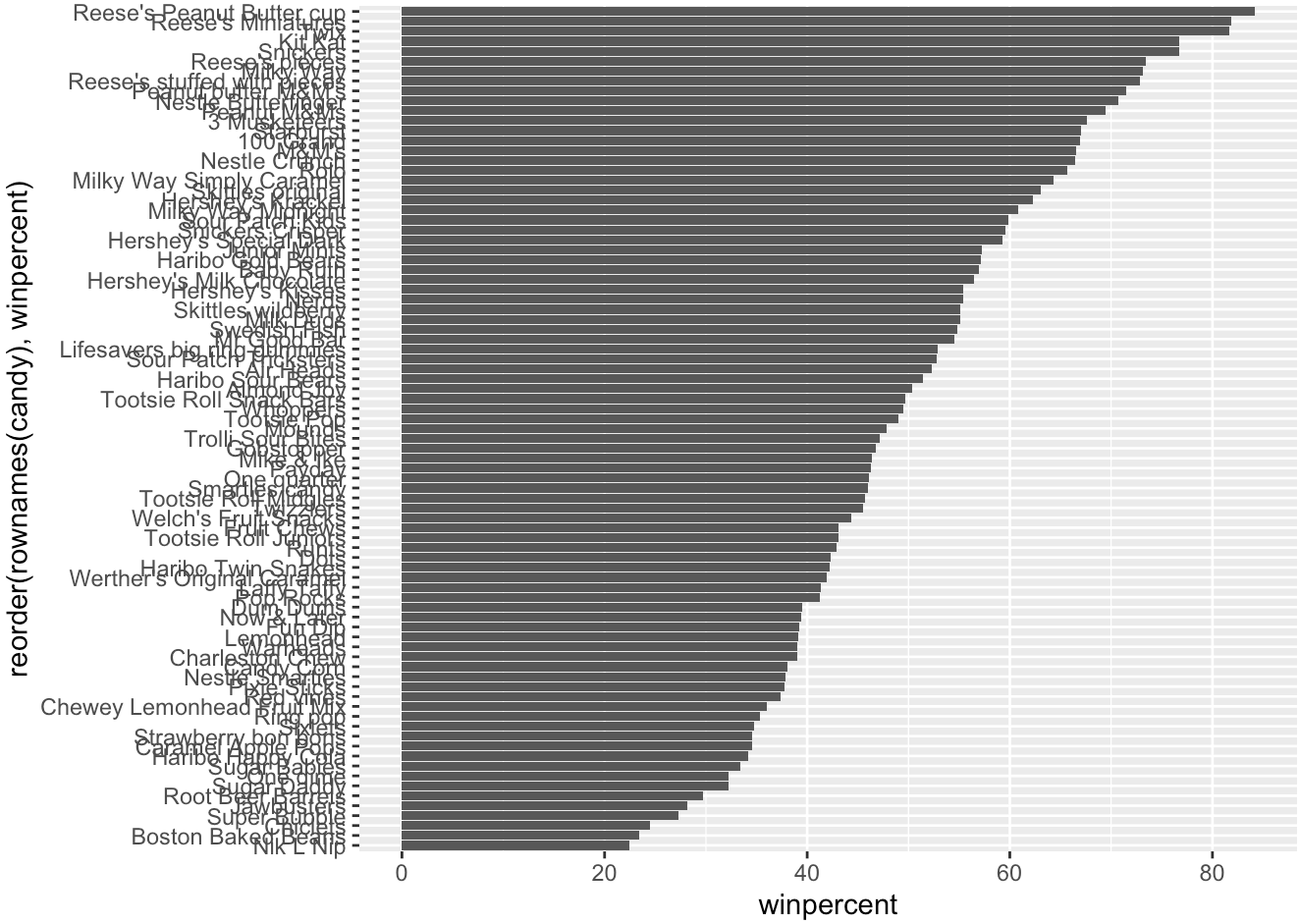
```
# head(candy[order(candy$winpercent),], n=5)
```

Q14. What are the top 5 all time favorite candy types out of this set?

```
# head(candy[order(candy$winpercent, decreasing=TRUE),], n=5)
```

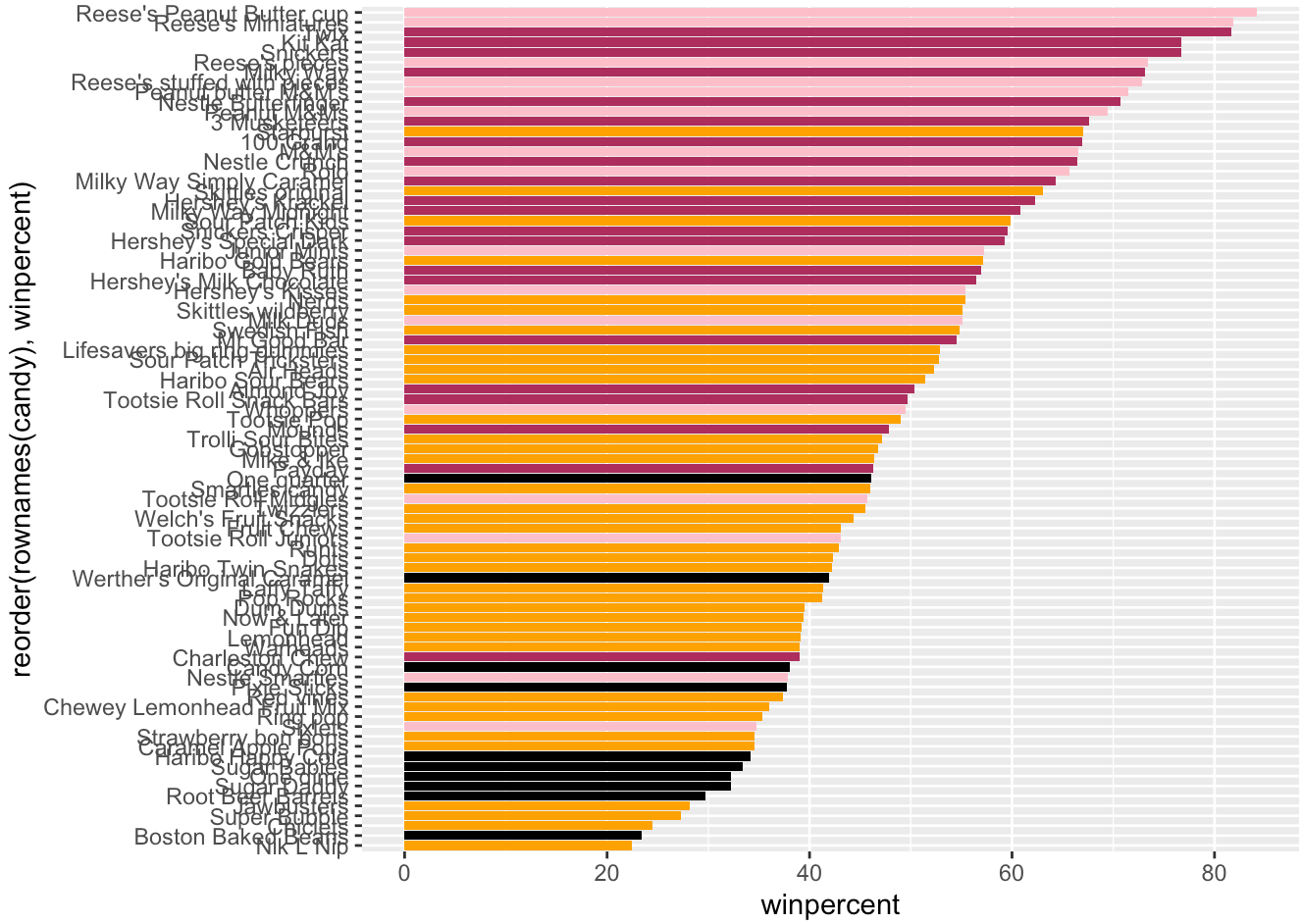
Q15. Make a first barplot of candy ranking based on winpercent values.

```
library(ggplot2)

ggplot(candy) +
  aes(winpercent, reorder(rownames(candy), winpercent)) +
  geom_col()
```



Let's setup a color vector that signifies candy type.

```
my_cols=rep("black", nrow(candy))
my_cols[as.logical(candy$chocolate)] = "pink"
my_cols[as.logical(candy$bar)] = "maroon"
my_cols[as.logical(candy$fruity)] = "orange"

ggplot(candy) +
  aes(winpercent, reorder(rownames(candy),winpercent)) +
  geom_col(fill=my_cols)
```



Now, for the first time, using this plot we can answer questions like:

Q17. What is the worst ranked chocolate candy?

```
# head(candy[order(candy$winpercent),], n=1)
```

Q18. What is the best ranked fruity candy?

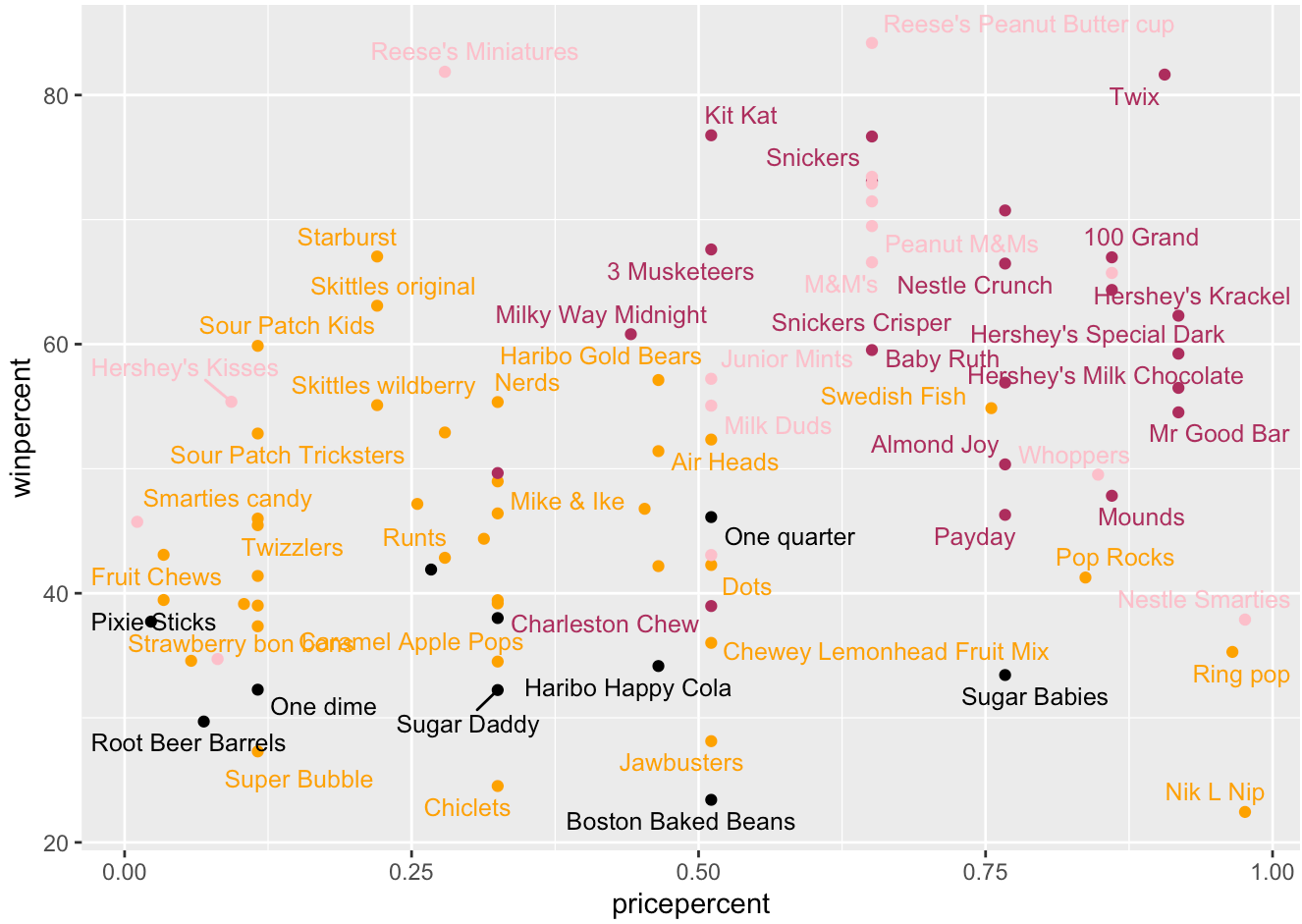
```
fruity_candy <- candy[as.logical(candy$fruity),]
fruity_candy[which.max(fruity_candy$winpercent), ]
```

```
##           chocolate fruity caramel peanutyalmondy nougat crispedricewafer hard
## Starburst         0      1       0              0      0                0    0
##           bar pluribus sugarpercent pricepercent winpercent
## Starburst   0        1        0.151         0.22   67.03763
```

The pricepercent variable records the percentile rank of the candy's price against all the other candies in the dataset. Lower values are less expensive and higher values are more expensive.

```
# install.packages("ggrepel")
library(ggrepel)

ggplot(candy) +
  aes(x = pricepercent,
      y = winpercent, label=rownames(candy)) +
  geom_point(col=my_cols) +
  geom_text_repel(col=my_cols, size=3.3, max.overlaps = 5)
```

```
## Warning: ggrepel: 27 unlabeled data points (too many overlaps). Consider
## increasing max.overlaps
```



Q19. Which candy type is the highest ranked in terms of winpercent for the least money - i.e. offers the most bang for your buck?

```
ord <- candy[order(candy$pricepercent, -candy$winpercent), ]
head( ord, 1 )
```

```
##                      chocolate fruity caramel peanutyalmondy nougat
## Tootsie Roll Midgies         1      0       0              0      0
##                      crispedricewafer hard bar pluribus sugarpercent
## Tootsie Roll Midgies                0    0   0        1        0.174
##                      pricepercent winpercent
## Tootsie Roll Midgies        0.011   45.73675
```

Q20. What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

```
most_expensive <- candy[order(-candy$pricepercent), ]

head(most_expensive, 5)
```

```
##                          chocolate fruity caramel peanutyalmondy nougat
## Nik L Nip                        0      1       0              0      0
## Nestle Smarties                  1      0       0              0      0
## Ring pop                         0      1       0              0      0
## Hershey's Krackel                1      0       0              0      0
## Hershey's Milk Chocolate         1      0       0              0      0
##                          crispedricewafer hard bar pluribus sugarpercent
## Nik L Nip                                0    0   0        1        0.197
## Nestle Smarties                          0    0   0        1        0.267
## Ring pop                                 0    1   0        0        0.732
## Hershey's Krackel                        1    0   1        0        0.430
## Hershey's Milk Chocolate                 0    0   1        0        0.430
##                          pricepercent winpercent
## Nik L Nip                       0.976   22.44534
## Nestle Smarties                 0.976   37.88719
## Ring pop                        0.965   35.29076
## Hershey's Krackel               0.918   62.28448
## Hershey's Milk Chocolate        0.918   56.49050
```

```
top5_expensive <- head(most_expensive, 5)
```

```
least_pop_expensive <- top5_expensive[order(top5_expensive$winpercent), ]

least_pop_expensive[1, ]
```

```
##           chocolate fruity caramel peanutyalmondy nougat crispedricewafer hard
## Nik L Nip         0      1       0              0      0                0    0
##           bar pluribus sugarpercent pricepercent winpercent
## Nik L Nip   0        1        0.197        0.976   22.44534
```

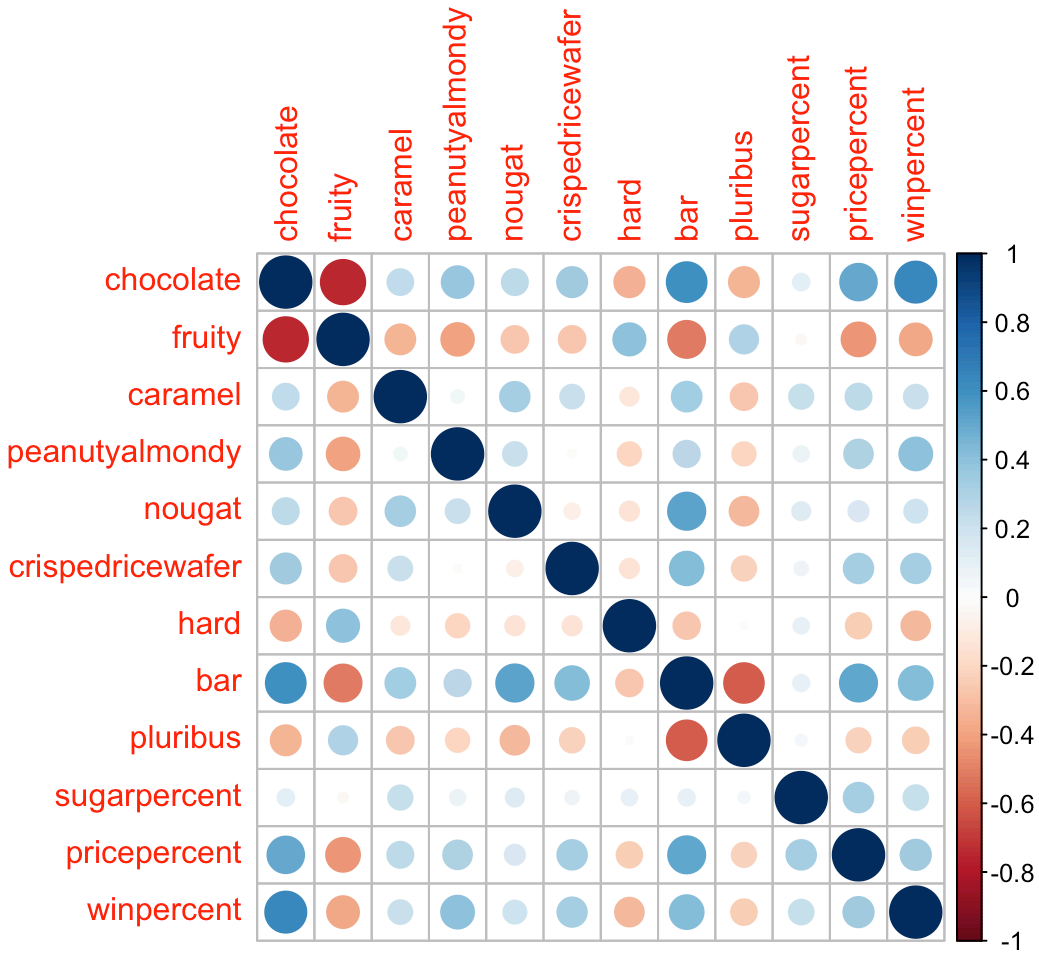
Q21. Optional…

Now that we've explored the dataset a little, we'll see how the variables interact with one another. We'll use correlation and view the results with the corrplot package to plot a correlation matrix.

```
# install.packages("corrplot")
library(corrplot)
```

```
## corrplot 0.95 loaded
```

```
cij <- cor(candy)
corrplot(cij)
```



Q22. Examining this plot what two variables are anti-correlated (i.e. have minus values)? *Fruity and chocolate candies*

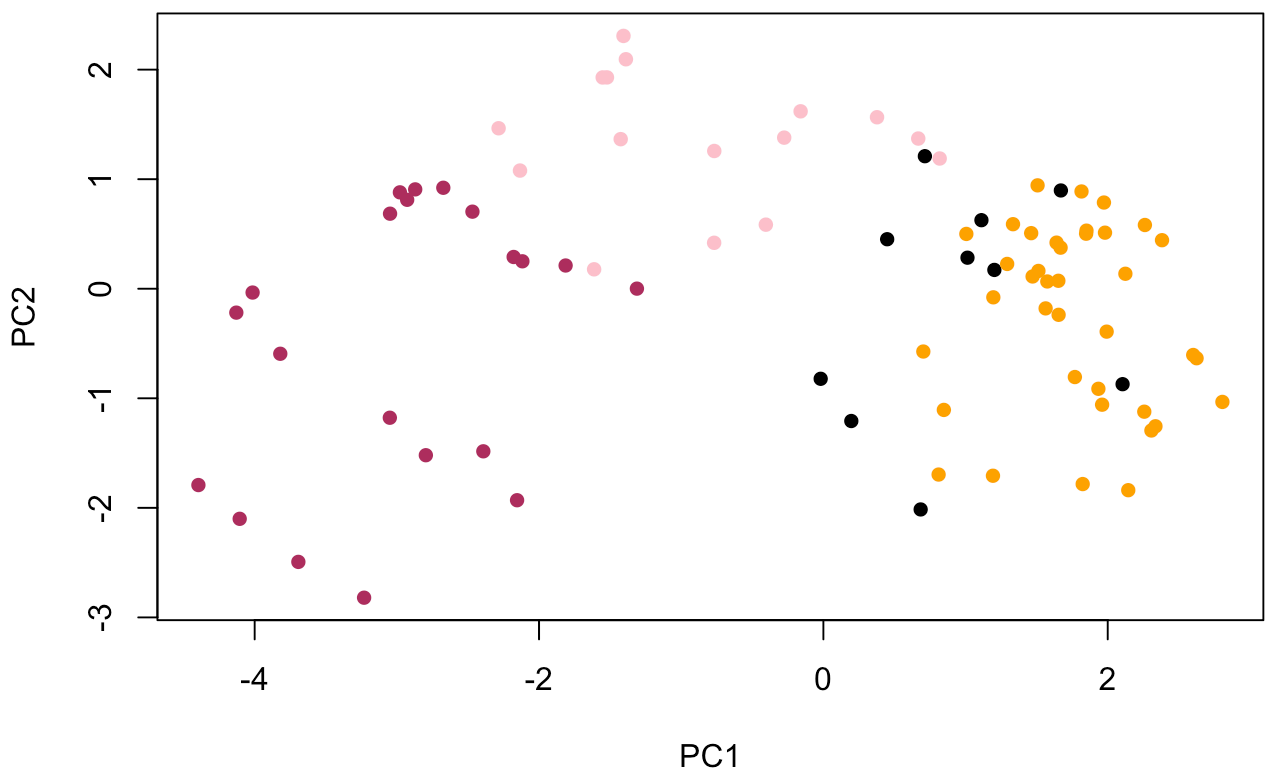
Q23. Similarly, what two variables are most positively correlated? *Chocolate and peanutyalmondy*

```
pca <- prcomp(candy, scale = TRUE)
summary(pca)
```

```
## Importance of components:
##                           PC1    PC2    PC3     PC4    PC5     PC6     PC7
## Standard deviation     2.0788 1.1378 1.1092 1.07533 0.9518 0.81923 0.81530
## Proportion of Variance 0.3601 0.1079 0.1025 0.09636 0.0755 0.05593 0.05539
## Cumulative Proportion  0.3601 0.4680 0.5705 0.66688 0.7424 0.79830 0.85369
##                            PC8     PC9    PC10    PC11    PC12
## Standard deviation     0.74530 0.67824 0.62349 0.43974 0.39760
## Proportion of Variance 0.04629 0.03833 0.03239 0.01611 0.01317
## Cumulative Proportion  0.89998 0.93832 0.97071 0.98683 1.00000
```

Q24. Complete the code to generate the loadings plot above. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you? Where did you see this relationship highlighted previously?

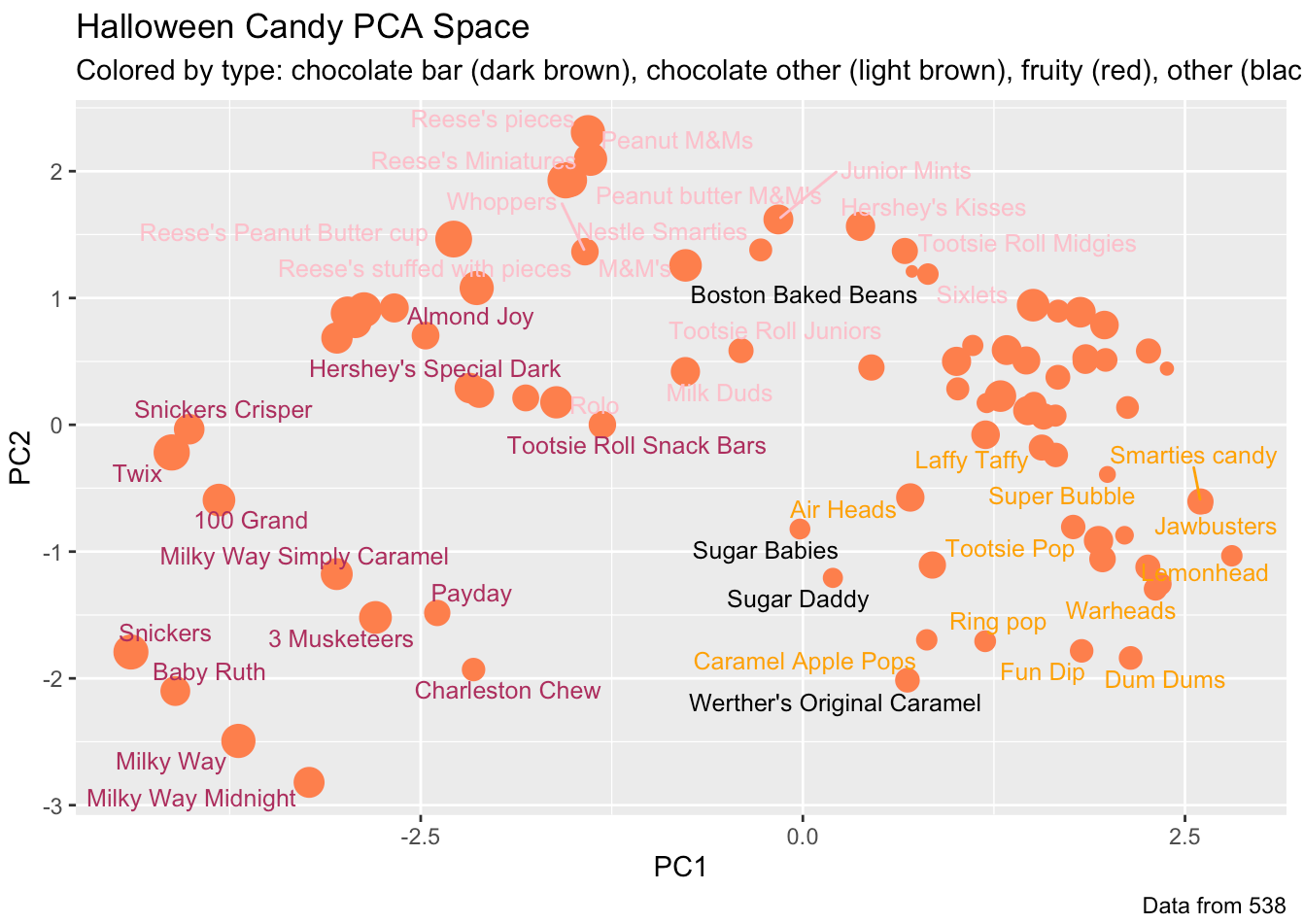
```
plot(pca$x[, 1:2], col=my_cols, pch=16)
```



```
my_data <- cbind(candy, pca$x[,1:3])

p <- ggplot(my_data) +
        aes(x=PC1, y=PC2,
            size=winpercent/100,
            text=rownames(my_data),
            label=rownames(my_data)) +
        geom_point(col="coral")
```

```
library(ggrepel)

p + geom_text_repel(size=3.3, col=my_cols, max.overlaps = 7)  +
  theme(legend.position = "none") +
  labs(title="Halloween Candy PCA Space",
       subtitle="Colored by type: chocolate bar (dark brown), chocolate other (light bro
wn), fruity (red), other (black)",
       caption="Data from 538")
```

```
## Warning: ggrepel: 39 unlabeled data points (too many overlaps). Consider
## increasing max.overlaps
```



```
# install.packages("plotly")

library(plotly)
```

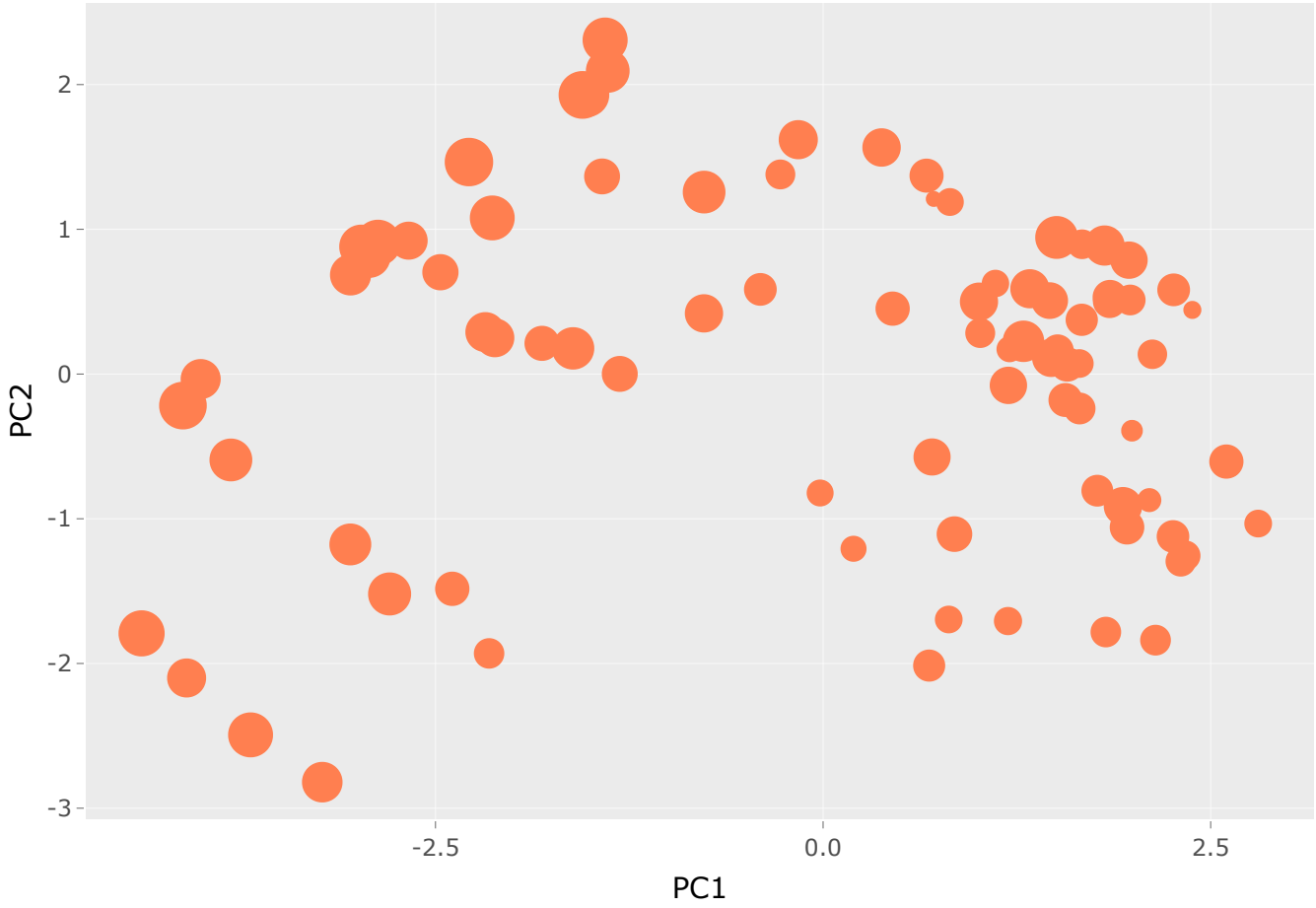
```
##
## Attaching package: 'plotly'
```

```
## The following object is masked from 'package:ggplot2':
##
##     last_plot
```

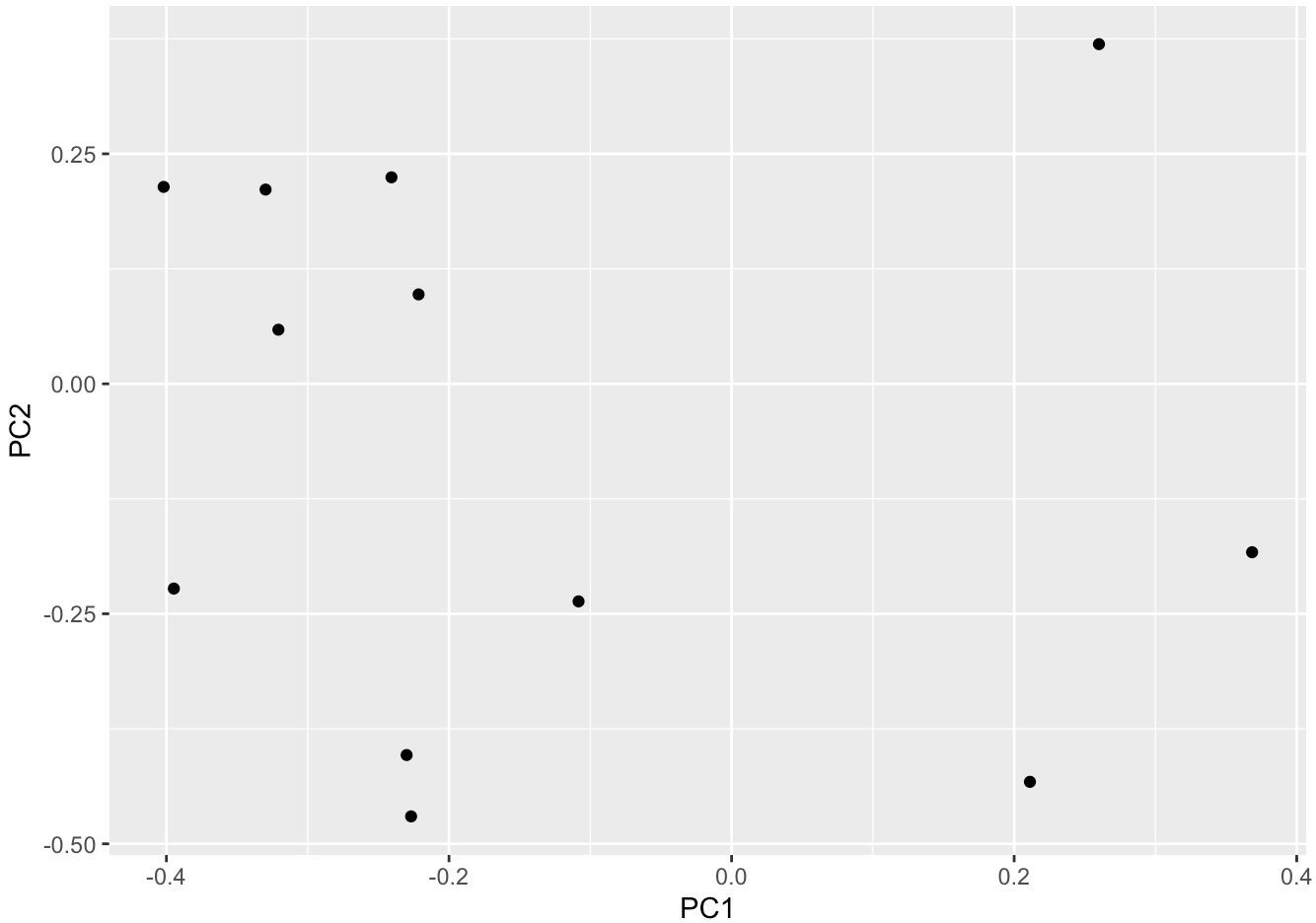
```
## The following object is masked from 'package:stats':
##
##     filter
```

```
## The following object is masked from 'package:graphics':
##
##     layout
```

```
ggplotly(p)
```



```
ggplot(pca$rotation) +
  aes(x = PC1, y = PC2) +
  geom_point()
```



Q25. Based on your exploratory analysis, correlation findings, and PCA results, what combination of characteristics appears to make a “winning” candy? How do these different analyses (visualization, correlation, PCA) support or complement each other in reaching this conclusion? *Overall, a “winning” candy tends to be chocolate-based, often with peanut/almond or nougat, and not hard or strongly fruity. Visualizations showed that these candies consistently rank higher in winpercent. Correlation analysis confirmed positive relationships between chocolate-related features and popularity, while fruity and hard candies were less favored. PCA tied these results together by showing that chocolate, nutty ingredients, and winpercent drive the main variation in the data, reinforcing the same conclusion from multiple perspectives.*